Kept: a guarded next best action for promises customers can keep

Team Sharpe Shooters · final architecture, 4 Oct 2026 · orange border = changed from the 2 Oct design · numbers from the real-data runs

31 source tables · CRM, cards, loans, deposits, cases, contacts, notes, transcripts, bureau, policy
five systems, five customer keys, no shared ID · 1M people · 77M records

1 · DATA PRODUCT FACTORY DuckDB · dbt · Splink · Soda

Bronze → Silver
land raw (all text) · tested cleaning rules
never guess: unreadable → quarantine

Identity resolution
exact keys (+ core-banking CIF) → Splink
≥0.95 auto · 0.80–0.95 steward
3.84M keys → 1,052,300 people

Golden C360
lineage per field · match confidence
hard stops · restricted table apart

Contract gate
ODCS from DC-COLL-001 · Q1–Q5
PASS · human review 99.3% (300)
schema drift: `kept map`

gold + silver + redacted text

C360

hardship status

2 · COLLECTIONS ASK LiteLLM (Gemini free tier) · sqlglot · LanceDB

Scope check
rules first (no LLM): protected, bulk PII, writes, other cases
+ classifier · "Try: …" on refusal

SQL ladder
certified metrics + reviewed templates
→ free SQL last (3 drafts, majority, worked examples) · dates in code

SQL guard
sqlglot AST: SELECT only,
17 tables, no PII columns,
row limit, 10 s timeout

Grounded answer
SQL / sources shown · numbers only from rows · policy, incidents, calls cited
dev 94% · refusals 100% / 100%

3 · TEXT + FEATURE STORE Presidio · Ollama · SetFit · dbt · Feast

Presidio redaction
EN + FR · SIN, postal, cards
785,594 texts · 0 leaks

Local LLM extraction
Gemma 3 12B on our GPU
verbatim quote per value
+ call summaries + QA

SetFit backfill
1,604 LLM labels → all
785K historical texts
hardship recall 0.98 / 0.99

Hardship signals
text + cash-flow rule +
agent · cease counts only
once an agent confirms

SQL features
DPD, promises, payday,
contacts, cohorts
26 features · point in time

Feature store (Feast)
same values train + live
parity 0 of 56,000
drift monitored (PSI)

features as of decision time

agent adds a note → live extraction (~5 s) → the card re-checks

4 · KEPT: GUARDED NEXT BEST ACTION rules (no AI) → model → arithmetic · LightGBM · MLflow · FastAPI · Next.js

1 Hard stops
cease, insolvency,
bankruptcy, deceased

2 Permissions
consent · local hours
3 contacts / 7 days

3 Hardship override
possible / confirmed or
vulnerable → protective only

4 Uplift model
LightGBM T-learner
Qini 0.0073 vs 0.0017

5–6 Kept sizing + rank
due 1–2 days after payday
within spare cash · capacity

7 Reasons + release
TreeSHAP top 3 · tiers
Red: specialist + 2nd

NBA card + app
96,000 cards / night
queue · Ask · steward · audit

Steps 1–3 and sizing are rules, no AI. The model never sends anything: a person releases every proposal, and the release re-checks every rule on fresh data.

Governance on every layer: role-based access · PII redacted before any LLM · free LLMs only, every call audited · no protected attributes or proxies (lineage CI test + curated tests on the trained model) · hardship protects, never scores · append-only audit · fairness report · drift + schema monitoring · 5% hold-out + kill switch